



Supporting Online Material for

Glass Transition Dynamics and Surface Layer Mobility in Unentangled Polystyrene Films

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Glass Transition Dynamics and Surface Layer Mobility in Unentangled Polystyrene Films

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Materials and Methods

1. Sample Preparation

Polystyrene (PS), with a weight-average molecular weight, M_w , of 2.4 kg/mol and polydispersity index of 1.06, was purchased from Scientific Polymer Products (Ontario, NY). To make the polymer films, we dissolve the PS granules in toluene and filter the resultant solution through a PTFE filter with nominal pore size of 0.1 μm (Fisher Scientific Co.) before spin-coating it onto the substrates to form the films. The substrates, purchased from Siltronix (France), are single crystal (001) silicon wafers covered by a 102 ± 5 nm thick thermal oxide. Before use, the wafers are cut into $1.5 \text{ cm} \times 1.5 \text{ cm}$ slides and submerged in a piranha solution (a mixture of sulfuric acid and hydrogen peroxide in 7:3 volume ratio) at 130°C for 10 min, followed by thorough rinsing in deionized water and drying by 99.99% nitrogen. Afterwards, the slides are further cleaned in oxygen plasma for 20 min whereupon ready for use. The thickness of the polymer film is controlled by either adjusting the concentration of the polymer solution or spinning speed, and monitored by ellipsometry at 5 different locations on the film. Deviation found in the measured thickness is less than 0.3 nm.

Most of the films were not annealed prior to measurement. We have found that pre-annealing has indiscernible effect on the measured viscosity (to be detailed below). During the viscosity measurement, the films are placed on a temperature-controlled hot plate inside a sealed chamber filled with 99.99% nitrogen at a slight positive pressure. The temperature is monitored by a thermocouple secured to a silicon wafer placed side by side the sample film. We find that the reading of the thermocouple can vary with its location on the hot plate within ± 5 cm from the center but not so much over time, with the amount of variation in the thermocouple reading increasing with increasing temperature. The error bars in Fig. 2A illustrate the maximum extent by which the thermocouple reading had varied.

2. Method to Measure the Viscosity

Previous studies showed that PS films on oxide coated silicon are unstable (S1, S2) and rupture into polymer beads upon heating (S2-S7). To determine the viscosity of the films, η , we follow the method detailed in Refs. (S8) and (S9). In brief, time-sequenced topographic images of the films are captured at various annealing times, t , by tapping-mode atomic force microscopy (AFM) as the films rupture. To convert the topographic data into power spectral density (PSD), $A^2_q(t)$, we first multiply the data by a Welch function before performing Fourier transformation and radial averaging, which results in the PSD, as those shown in Fig. 1E (S5). In this experiment, a local height fluctuation, $\delta h(\mathbf{r}, t)$, in the film produces a local excess pressure, $P(\mathbf{r}, t) = -\gamma \nabla^2 \delta h(\mathbf{r}, t) + G''(h) \delta h(\mathbf{r}, t)$ (S7, S10), where $\mathbf{r}$ is a position vector in the plane of the film, γ is the surface tension of the polymer and $G(h)$ is the van der Waals potential of the film. The excess pressure $P(\mathbf{r}, t)$ induces a two-dimensional polymer (flow) flux in

the film. In the long-wavelength limit, $qh \ll 1$, valid in this experiment, the planar flux is given by:

$$j(\mathbf{r}, t) = -M_{\text{tot}} \nabla P(\mathbf{r}, t) + \xi(\mathbf{r}, t),$$

where M_{tot} is the mobility of the film, defined by $[\int_0^h v(z) dz]/(-\nabla P)$ (S7, S10, S11), with $v(z)$ being the fluid velocity profile in the film instigated by ∇P , and $\xi(\mathbf{r}, t)$ denotes conserved noise arising from thermal activations. A linear stability analysis on the associated flow continuity equation predicts the following expression for the time-dependent PSD (S7, S10):

$$A_q^2(t) = A_q^2(0) \exp(2\Gamma_q t) + \left[\frac{k_B T}{G''(h) + \gamma q^2} \right] (1 - \exp(2\Gamma_q t)), \quad (\text{S1a})$$

$$\text{where} \quad \Gamma(q) = -M_{\text{tot}} [G''(h) q^2 + \gamma q^4], \quad (\text{S1b})$$

and k_B is the Boltzmann constant and T is absolute temperature. We have previously shown (S7-S9) that eqn. (S1a,b) describes the evolution of the surface of spin-coated polymer films very well. If the film is homogeneous, i.e., both its thickness h and viscosity η are uniform, and the no-slip boundary condition applies at the bottom interface, $M_{\text{tot}} = h^3/3\eta$. For generalization of M_{tot} to layered structures, please refer to the main text. To determine the viscosity of a film, we fit the $A_q^2(t)$ data of the film to eqn. (S1a,b) by using the viscosity parameter, η , as the only fitting parameter while keeping the other parameters fixed to the published and/or experimental values, after accounting for a low- q spectral background (S8). In performing the fits, we use only the data from the initial stage of roughening in the linear evolution regime well before the formation of any holes.

The first term on the right hand side of Equation (S1a) accounts for the conventional spinodal growth of the initial roughness of the film, $A_q^2(0)$. The second term accounts for thermal noise induced growth of the surface fluctuations. If the first term dominates, eqn. (S1a) reduces to the expression, $A_q^2(t) = A_q^2(0)\exp(2\Gamma(q)t)$, commonly used to describe the rupturing process of an unstable film (S4, S10, S12). Correspondingly, the PSD of the film, $A_q^2(t)$, would be peaked at the constant wavevector, $q_m \equiv \sqrt{-G''(h)/(2\gamma)}$, which maximizes $\Gamma(q)$ (n.b. $G''(h) < 0$ for unstable films). For the rupturing of polystyrene films on silicon, we have shown that the thermal noise induced fluctuations are significant (S9). From eqn. (S1a,b), the thermal noise induced fluctuations emerge systematically from high to low wavevectors. This explains the initial shift of the PSD peak seen in Fig. 1E and its subsequent convergence to $\sim 16 \mu\text{m}^{-1}$, which coincides with the estimated value of q_m .

Figure S1 shows that our method provides a sensitive measure for η . For a set of PSD's obtained from a $h = 4$ nm film annealed at 334 K, the model fit becomes visibly degraded when the value of η is adjusted by $\pm 12\%$ from the optimal value of 5.3×10^5 Pas. As a result, the value of the chi-square parameter increases by 9 to 26%. The error bars shown in Figs. 2A and 3 for the viscosity measurements represent the deviation from the optimal value of η that causes the quality of the fits to degrade visibly. It is remarkable that all the data, including those of the $h = 2.3$ nm films fit well to eqn. (S1a,b) even though 2.3 nm is comparable to the radius of gyration of the polymer (~ 1.3 nm). Since our model has assumed continuum hydrodynamics, this finding suggests that polymer layers with thickness down to ~ 2 nm can still be approximated by the continuum limit.

Previous studies had found that pre-annealing the films can have a significant influence on their dynamical behaviors (S13-S16). We have examined if pre-

annealing may affect our measurements by annealing several 4 nm films at 323 K, i.e., 32 K above the T_g of the films for 30 minutes in nitrogen before measuring their viscosity. The data obtained from one of the films measured at 334 K is shown in Fig. S2. The result shows that the annealed films give the same viscosity as the unannealed ones within experimental uncertainty. This is in keeping with our observation that the viscosities of the 79 nm films shown in Fig. 2A (which were not pre-annealed) are the same as the published data of the bulk polymer. This is also consistent with the fact that only a single value of viscosity is required to model the data obtained in all of our experiments, including those performed near or below the T_g of the bulk polymer, where the measurement time usually exceeds two hours (please see Figs. 1E and S2). These show that the majority of the structural relaxations, if any, in the films that may be caused by pre-annealing or annealing en route the experiment occur within an insignificant portion of the measurement time. Such a short structural relaxation time may arise from the polymer used here being made of short, unentangled chains, different from those used in the other studies (S13-S16).

3. Method to Determine T_K and its Error

The value of T_K is determined by least-square fitting the data of Fig. 2A to the Vogel-Fulcher-Tammann (VFT) relation (eqn. (1)):

$$\eta(T) = \eta(\infty) \exp\left(\frac{B}{T - T_K}\right),$$

where T_K and $\eta(\infty)$ are taken as the fitting parameters and B fixed to the bulk value of 1620 K (S17). The errors of T_K shown in Fig. 2A are determined by fixing $\eta(\infty)$ to the

best-fitted value while finding the maximum extent over which the value of T_K can be changed and still produce a model line that can pass through all the data points within their error brackets.

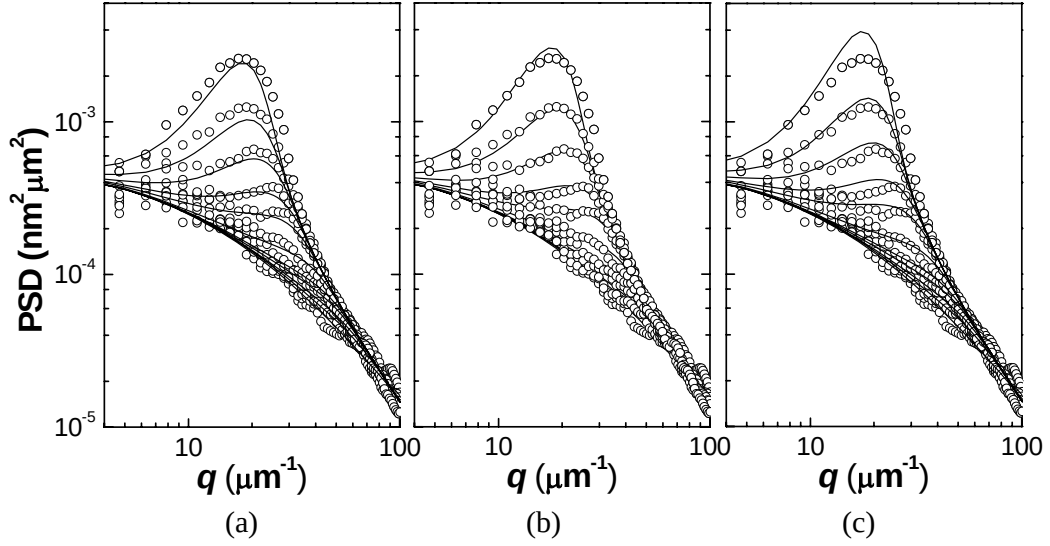


Fig. S1. Illustration of the sensitivity of the model fitting to variations in the value of η . The open circles represent the power spectral density, PSD, of a film with thickness, $h = 4$ nm after annealing at 334 K for times (from bottom to top) = 0, 15, 30, 60, 90, 210, 600, 1080, 2100, 3840 and 7200 s. The solid lines are calculations by using eqn. (S1a,b) with different values of η : (a) 6×10^5 Pas, (b) 5.3×10^5 Pas (optimal), and (c) 4.6×10^5 Pas; the respective values of chi-square are: 1.24×10^{-5} , 1.14×10^{-5} , and 1.44×10^{-5} . The quality of the fit worsens noticeably when η is adjusted by less than 15% from the optimal value.

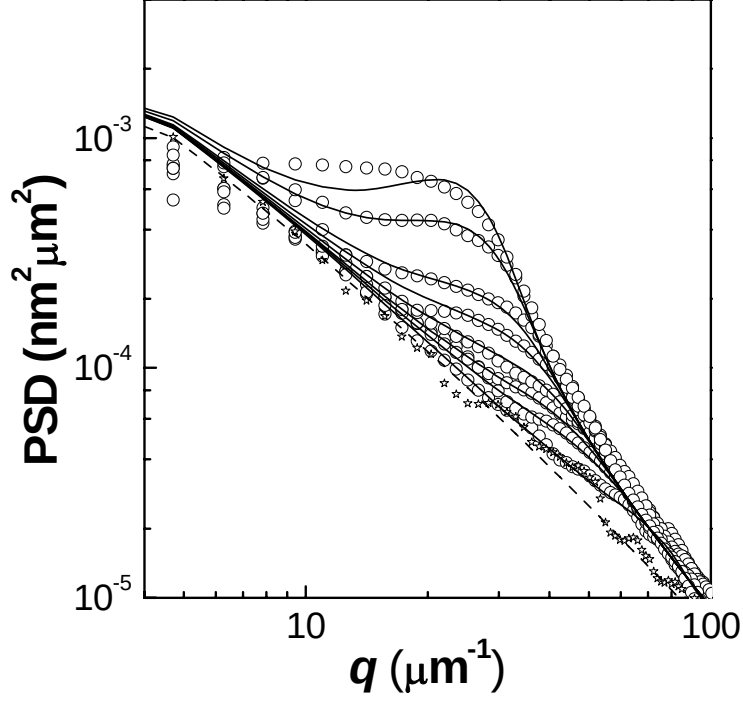


Fig. S2. Power spectral density obtained from a film with thickness, $h = 4$ nm, first pre-annealed at 323 K for 1800 s before subsequent time-sequenced measurement at 334 K. The stars are the data of the as-cast film. The dash line is to guide the eye. The open circles represent the data obtained upon heating at 334 K for times (from bottom to top) = 0, 20, 60, 120, 240, 480, 1080, 2100 s. The solid lines are the best fitted model lines. The fitted value of η here, being $(5 \pm 1) \times 10^5$ Pas agrees with the value of $(5.3 \pm 0.7) \times 10^5$ Pas obtained from the film not pre-annealed (Fig. S1 and 1E).

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